Project Coversheet

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Project Title (Example – Week1, Week2, Week3, Week 4)	Week 3 - Project: Churn Prediction for StreamWorks Media.

Instructions:

Students must download this cover sheet, use it as the first page of their project, and then save the entire document as a PDF before submission.

Project Guidelines and Rules

1. Formatting and Submission

- Format: Use a readable font (e.g., Arial/Times New Roman), size 12, 1.5 line spacing.
- Title: Include Week and Title (Example - Week 1: Travel Ease Case Study.)
- File Format: Submit as PDF or Word file
- Page Limit: 4–5 pages, including the title and references.

2. Answer Requirements

- Word Count: Each answer should be within 100–150 words; Maximum 800–1,200 words.
- Clarity: Write concise, structured answers with key points.
- Tone: Use formal, professional language.

3. Content Rules

- Answer all questions thoroughly, referencing case study concepts.

- Use examples where possible (e.g., risk assessment techniques).
- Break complex answers into bullet points or lists.

4. Plagiarism Policy

- Submit original work; no copy-pasting.
- Cite external material in a consistent format (e.g., APA, MLA).

5. Evaluation Criteria

- Understanding: Clear grasp of business analysis principles.
- Application: Effective use of concepts like cost-benefit analysis and Agile/Waterfall.
- Clarity: Logical, well-structured responses.
- Creativity: Innovative problem-solving and examples.
- Completeness: Answer all questions within the word limit.

6. Deadlines and Late Submissions

- Deadline: Submit on time; trainees who fail to submit the project will miss the "Certificate of Excellence"

7. Additional Resources

- Refer to lecture notes and recommended readings.
- Contact the instructor or peers for clarifications before the deadline.

YOU CAN START YOUR PROJECT FROM HERE

Business Questions to Answer

1. Do users who receive promotions churn less?

The Chi-square test showed $p = 0.138$ (> 0.05), indicating that receiving promotions is not significantly associated with churn. The promotions bar chart also shows similar churn patterns between promoted and non-promoted users.

2. Does watch time impact churn likelihood?

The independent t-test produced $p = 0.852$ (> 0.05), meaning there is no statistically significant difference in average watch hours between churned and retained users.

3. Are mobile dominant users more likely to cancel?

The correlation between mobile app usage and churn is weak, and Logistic Regression assigns only a small coefficient to this variable. Mobile usage alone is therefore not a reliable predictor of churn.

4. What are the top 3 features influencing churn based on your model?

- Based on the Logistic Regression model:
  1. is_loyal
  2. country_UK
  3. country_France
- These variables have the largest positive coefficients and therefore contribute most strongly to the model's churn predictions.

5. Which customer segments should the retention team prioritize?

Based on the exploratory analysis and the Logistic Regression model, the retention team should focus on:

- Users with low loyalty (short tenure), as loyalty is the strongest predictor in the model.
- Customers from the UK and France, since these regional variables have the highest model coefficients.
- Users with lower engagement levels, as they may be more likely to become inactive over time.

By targeting these customer groups with personalized communication and retention campaigns, StreamWorks Media can reduce churn and improve long-term customer value.

6. What factors affect user watch time or tenure? (Linear regression insight)

The Linear Regression model shows that the available customer features explain only a small proportion of watch time variation ($R^2 = -0.012$). This suggests that additional variables such as content preferences, viewing frequency and user engagement would be required to build a more accurate model.

7. **Recommendations:** Suggest 2-3 actionable strategies

- Focus retention efforts on customer engagement and loyalty rather than demographic characteristics.
- Collect richer behavioural data (viewing frequency, favourite genres, login patterns) to improve churn prediction.
- Test more advanced machine learning models such as Random Forest or Gradient Boosting and address class imbalance to improve churn detection performance.

8. Data Issues or Risks

Mention any limitations or risks (e.g., data imbalance, feature leakage)

Note: Include output screenshots from your jupyter notebook where required.

- The dataset contains considerably more retained users than churned users. As a result, the Logistic Regression model predicts almost every customer as retained, leading to high accuracy but poor churn detection.
- The dataset contains basic demographic and subscription information but lacks detailed viewing behaviour and customer interaction data.
- The Logistic Regression model achieved an ROC-AUC score close to 0.50 and the Linear Regression model produced a negative R^2 score, indicating limited predictive power.